

Feature Dictionary

Customer Personality Segmentation Project

Module 05 — Feature Engineering & Data Preprocessing (CPS-M05)

1. Purpose

This document defines every behavioural feature engineered from the cleaned Customer Personality Analysis dataset, including its formula, purpose, business significance, and expected impact on customer segmentation (clustering).

2. Feature Dictionary

Feature Name	Formula	Purpose	Business Significance	Expected Impact
Customer_Age	Reference_Year - Year_Birth	Captures customer's current age.	Relates to income stability, family stage, product preference.	Separates younger vs older segments.
Customer_Tenure	Reference_Date - Dt_Customer (days)	How long the customer has been enrolled.	Longer-tenured customers may be more loyal.	Distinguishes long-term vs new customers.
Total_Children	Kidhome + Teenhome	Combined dependents count.	Household composition affects discretionary spend.	Separates family-oriented from single/couple shoppers.
Family_Size	1 + (1 if Married/Together) + Total_Children	Estimated household size.	Larger families allocate budgets differently.	Segments by household scale.
Total_Spending	Sum of all Mnt* columns	Overall spend across categories.	Direct measure of customer value.	Core driver of high vs low-value segments.
Total_Purchases	Web+Catalog+Store+Deals purchases	Total transaction count.	Indicates purchase frequency.	Identifies frequent vs occasional buyers.
Total_Campaign_Acceptance	Sum of AcceptedCmp1-5 + Response	Campaigns accepted (0-6).	Reflects marketing receptiveness.	Separates responders from non-responders.
Avg_Spending_Per_Purchase	Total_Spending / Total_Purchases	Average basket value.	Distinguishes big-basket vs frequent-small-basket buyers.	Refines value segmentation.
Digital_Engagement	NumWebVisitsMonth + NumWebPurchases	Combined online activity score.	Measures digital-channel activity.	Flags customers for digital campaigns.
Deal_Dependency	NumDealsPurchases / Total_Purchases	Share of deal-driven purchases.	High values = price-sensitive behaviour.	Separates discount seekers from loyal full-price buyers.
Preferred_Shopping_Channel	argmax(Web, Catalog, Store purchases)	Customer's top purchase channel.	Guides channel-specific marketing.	Enables channel-based segmentation.
Product_Preference	argmax(Mnt* columns)	Customer's top-spend product category.	Reveals product affinity.	Enables product-based cross-selling segments.
Customer_Activity_Level	Rule-based on Recency & Total_Purchases	High/Medium/Low engagement label.	Flags churn risk vs highly engaged.	Directly usable as an engagement segment label.

3. Notes

- Reference date used for tenure calculations: 30 July 2026.

- All 13 engineered features are derived purely from existing cleaned columns — no external data is introduced.
- Categorical engineered features (Preferred_Shopping_Channel, Product_Preference, Customer_Activity_Level) are further encoded in Task 2 for ML compatibility.